

theorem

Vera: Weakest Preconditions for TICL over Rust

From Imp Structural Lemmas to Rust VCGen Rules

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Background: TICL over Imp

TICL Structural Lemmas for Imp

Core idea (from the TICL paper)

For a small imperative language `StImp` with mutable state, the entailment $[t, m \Vdash_L \varphi]_S$ is proved structurally via inference rules on the syntax of t .

Key rules from `imp-lemmas.tex`:

- **Seq**_SAU_L: sequential composition decomposes into bind
- **If**_SAU_L: conditional branches both satisfy the obligation
- **While**_SAG: infinite loop with coinductive invariant $\mathcal{R}$
- **While**_SAU_L: terminating loop with ranking function f

Goal: Lift these to Rust syntax $\rightarrow$ define $\text{wp}(_, _)$ for Vera.

Notation

$\sigma \in \text{State}$ (mutable references, heap)

$[t, \sigma \Vdash_L \varphi]_{\text{Rust}} \triangleq \text{“Rust term } t \text{ from state } \sigma \text{ satisfies } \varphi\text{”}$

$\text{wp}(t, \varphi) \triangleq \text{weakest precondition on } \sigma \text{ s.t. } [t, \sigma \Vdash_L \varphi]_{\text{Rust}}$

Temporal formulas

- $\varphi ::= \text{now } P \mid \text{AG } \varphi \mid \varphi \text{ AU } \varphi' \mid \varphi \wedge \varphi'$
- Sugar: $\text{AF } \varphi' = \top \text{ AU } \varphi', \quad \text{AX } \varphi' = \top \text{ AN } \varphi'$
- done Q : termination postcondition; now P : intermediate state predicate

Inference Rules (Single-Process)

Assignment

Rust: $*x = e$

$$\frac{\begin{array}{c} [e, \sigma \Vdash_R \varphi \text{ AU done } \mathcal{R}]_{\text{Rust}} \\ \forall v, \sigma', \mathcal{R} \ v \ \sigma' \rightarrow [\text{skip}, \sigma'[x \mapsto v] \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}} \end{array}}{[*x = e, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{ASSIGN}_{\text{RUST}} \text{AU}_L$$

Evaluating e may take steps; φ must hold along the path.

Sequential Composition (let-binding)

Rust: `let x = t; u`

$$\frac{[t, \sigma \Vdash_R \varphi \text{ AU done } \mathcal{R}]_{\text{Rust}} \quad \forall x, \sigma', \mathcal{R} \ x \ \sigma' \rightarrow [u[x], \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}}{[\text{let } x = t; u, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{LET}_{\text{RUST}} \text{AU}_L$$

Direct lift of **BindAU**_L / **SeqS**AU_L from TIDL.

Rust: `if c {t} else {u}`

$$\frac{\begin{array}{l} [c, \sigma \Vdash_R \varphi \text{ AU done} = (b, \sigma')]_{\text{Rust}} \\ \left\{ \begin{array}{ll} [t, \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}, & \text{if } b \\ [u, \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}, & \text{otherwise} \end{array} \right. \end{array}}{[\text{if } c \{t\} \text{ else } \{u\}, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{IF}_{\text{RUST}}\text{AU}_L$$

Evaluating c may take multiple steps; φ must hold along the path.

While Loop (AU – terminating)

Rust: `while c {t}` **with** invariant $\mathcal{R}$, decreases f

$$\frac{\begin{array}{l} \mathcal{R} \sigma \\ \forall \sigma, \mathcal{R} \sigma \rightarrow [c, \sigma \Vdash_R \varphi \text{ AU done} = (b, \sigma')]_{\text{Rust}} \\ \wedge \left\{ \begin{array}{l} [t, \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}} \vee \\ [t, \sigma' \Vdash_R \varphi \text{ AU done } (\lambda \sigma''. \mathcal{R} \sigma'' \wedge f \sigma'' < f \sigma)]_{\text{Rust}}, \text{ if } b \\ \varphi'(\sigma'), \text{ otherwise} \end{array} \right. \end{array}}{[\text{while } c \{t\}, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{WHILE}_{\text{RUST}} \text{AU}_L$$

While Loop (AG – infinite)

Rust: `while c {t}` **with** invariant $\mathcal{R}$ (**no** decreases)

$$\frac{\begin{array}{c} \mathcal{R} \sigma \\ \forall \sigma, \mathcal{R} \sigma \rightarrow \\ [\text{while } c \{t\}, \sigma \Vdash_L \varphi]_{\text{Rust}} \wedge \\ [c, \sigma \Vdash_R \varphi \text{ AU done} = (\text{true}, \sigma)]_{\text{Rust}} \wedge \\ [t, \sigma \Vdash_R \varphi \text{ AU done } (\lambda \sigma' \Rightarrow \mathcal{R} \sigma')]_{\text{Rust}} \end{array}}{[\text{while } c \{t\}, \sigma \Vdash_L \text{AG } \varphi]_{\text{Rust}}} \text{WHILE}_{\text{RUST}} \text{AG}$$

Coinductive hypothesis $[\text{while } c \{t\}, \sigma \Vdash_L \varphi]_{\text{Rust}}$; condition must always be true.

Function Call (Bind Rule)

Rust: `let x = f( $\vec{e}$ ); k x`

$$\frac{\begin{array}{c} [f(\vec{e}), \sigma \Vdash_R \varphi \text{ AU done } \mathcal{R}]_{\text{Rust}} \\ \forall x, \sigma', \mathcal{R} \ x \ \sigma' \rightarrow [k \ x, \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}} \end{array}}{[\text{let } x = f(\vec{e}); k \ x, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{CALL}_{\text{RUST}} \text{AU}_L$$

Instance of Let rule. Call is a sub-expression that produces a value.

Single-Process Weakest Preconditions

$$\begin{aligned} \text{wp}(*x = e, \varphi \text{ AU } \varphi') &= \lambda \sigma. \text{wp}(e, \varphi \text{ AU done } \mathcal{R}) \sigma \\ &\quad \wedge \forall v, \sigma'. \mathcal{R} v \sigma' \rightarrow \varphi'(\sigma'[x \mapsto v]) \end{aligned}$$

wp: Let-binding (Sequential Composition)

$$\begin{aligned} \text{wp}(\text{let } x = t; u, \varphi \text{ AU } \varphi') &= \lambda \sigma. \text{wp}(t, \varphi \text{ AU done } \mathcal{R}) \sigma \\ &\quad \wedge \forall x, \sigma'. \mathcal{R} \ x \ \sigma' \rightarrow \text{wp}(u[x], \varphi \text{ AU } \varphi') \sigma' \end{aligned}$$

$$\text{wp}(\text{if } c \{t\} \text{ else } \{u\}, \varphi \text{ AU } \varphi') = \lambda \sigma.$$

$$\text{wp}(c, \varphi \text{ AU done } \mathcal{R}) \sigma \wedge \forall b, \sigma'. \mathcal{R} b \sigma' \rightarrow$$

$$\begin{cases} \text{wp}(t, \varphi \text{ AU } \varphi') \sigma', & \text{if } b \\ \text{wp}(u, \varphi \text{ AU } \varphi') \sigma', & \text{otherwise} \end{cases}$$

wp: While Loop (AU)

$$\text{wp}(\text{while } c \{t\}, \varphi \text{ AU } \varphi') = \lambda \sigma.$$

$$\mathcal{R} \sigma \wedge \forall \sigma, \mathcal{R} \sigma \rightarrow$$

$$\text{wp}(c, \varphi \text{ AU done } \mathcal{R}') \sigma \wedge \forall b, \sigma'. \mathcal{R}' b \sigma' \rightarrow$$

$$\begin{cases} \text{wp}(t, \varphi \text{ AU } \varphi') \sigma' \vee \\ \text{wp}(t, \varphi \text{ AU done } (\lambda _ \sigma''. \mathcal{R} \sigma'' \wedge f \sigma'' < f \sigma)) \sigma', & \text{if } b \\ \varphi'(\sigma'), & \text{o.w.} \end{cases}$$

wp: While Loop (AG)

$$\text{wp}(\text{while } c \{t\}, \text{AG } \varphi) = \lambda \sigma.$$

$$\mathcal{R} \sigma \wedge \forall \sigma, \mathcal{R} \sigma \rightarrow$$

$$\text{wp}(\text{while } c \{t\}, \varphi) \sigma \wedge$$

$$\text{wp}(c, \varphi \text{ AU done} = (\text{true}, \sigma)) \sigma \wedge$$

$$\text{wp}(t, \varphi \text{ AU done } (\lambda \sigma'. \mathcal{R} \sigma')) \sigma$$

Coinductive: $\text{wp}(\text{while } c \{t\}, \varphi) \sigma$ is the coinductive hypothesis.

wp: Function Call – f ensures AF done $\mathcal{R}_f$

f has terminating postcondition

$$\text{wp}(f(e_1, \dots, e_i), \varphi \text{ AU } \varphi') = \lambda \sigma. \forall x, \sigma'. \mathcal{R}_f x \sigma' \rightarrow \text{wp}(k x, \varphi \text{ AU } \varphi') \sigma'$$

where k is the continuation after the call.

f has temporal ensures Ψ : transparent checking

Additionally assert $\Psi \implies \Phi$ where Φ is the caller's obligation.

f ensures AG ψ : cannot prove AF for it

If f has AG ψ (diverges), then $\text{wp}(f(\vec{e}), \text{AF } \varphi')$ is **unprovable** – a diverging call can never reach the AF goal.

Multi-Process Configuration

Configuration

A **configuration** $C = (P, \sigma, i)$:

$P : \text{PID} \rightarrow \text{Term}$

process dictionary

$\sigma \in \text{State}$

shared mutable state

$i \in \text{PID}$

active process

- Cooperative scheduling: run until `.await`, then yield
- Temporal formulas φ are over shared state σ
- $\text{WP}(C, \varphi)$ – multi-process wp, defined by cases on $P(i)$

Multi-Process Weakest Preconditions

WP: Sequential Step

When $P(i)$ is a non-async construct (assignment, let, if, while, sync call):

$$\text{WP}((P, \sigma, i), \varphi) = \text{wp}(P(i), \varphi) \sigma$$

Delegate to single-process $\text{wp}(,)$. Process dictionary P is unchanged.

Programs without `async/await` reduce entirely to single-process wp :

$$\text{WP}((\{0 \mapsto t\}, \sigma, 0), \varphi) = \text{wp}(t, \varphi) \sigma$$

WP: Async Bind (Lazy Future Creation)

Rust: $P(i) = \text{let } p = \text{async } f(\vec{e}); k \ p$

$$\text{WP}((P, \sigma, i), \Phi) = \text{WP}((P[i \mapsto k, p \mapsto f_{body}], \sigma, i), \Phi)$$

- Creates new process p with body f_{body} (suspended)
- Replaces active process i 's term with continuation k
- No state change, no scheduling – futures are lazy

WP: Await (AU – terminating callee)

Rust: $P(i) = \text{let } x = p.\text{await}; k\ x$

$$\begin{aligned} \text{WP}((P, \sigma, i), \varphi \text{ AU } \varphi') = \\ \text{WP}((P, \sigma, p), \varphi \text{ AU done } \mathcal{R}) \\ \wedge \forall x, \sigma'. \mathcal{R}\ x\ \sigma' \rightarrow \text{WP}((P[i \mapsto k[x]], \sigma', i), \varphi \text{ AU } \varphi') \end{aligned}$$

- Switch active process to p ; run it maintaining φ
- After p terminates with result x and state σ' , resume i

WP: Await (AG – diverging callee)

Rust: $P(i) = p.\text{await}$ **where** $P(p)$ **satisfies** $\text{AG } \psi$

$$\text{WP}((P, \sigma, i), \text{AG } \varphi) = \text{WP}((P, \sigma, p), \text{AG } \psi) \wedge (\psi \implies \varphi)$$

- Process p runs forever (AG); await diverges
- Caller's $\text{AG } \varphi$ discharged if $\psi \implies \varphi$
- Continuation after await is unreachable

Summary

Complete Rule Set

Construct	Rule	Layer
$*x = e$	State update	$\text{wp}(,)$
$\text{let } x = t; u$	BindAU_L	$\text{wp}(,)$
$\text{if } c \{t\} \text{ else } \{u\}$	Case split (effectful c)	$\text{wp}(,)$
$\text{while } c \{t\} + \text{dec.}$	$\text{While}_{\text{Rust}}\text{AU}_L$	$\text{wp}(,)$
$\text{while } c \{t\} \text{ (no exit)}$	$\text{While}_{\text{Rust}}\text{AG}$	$\text{wp}(,)$
$f(\vec{e})$	Contract-dependent	$\text{wp}(,)$
Sequential step	Delegate to $\text{wp}(,)$	$\text{WP}(,)$
$\text{async } f(\vec{e})$	Extend P , no-op	$\text{WP}(,)$
$p.\text{await}$ (AU callee)	Switch active to p	$\text{WP}(,)$
$p.\text{await}$ (AG callee)	Divergence	$\text{WP}(,)$

Two layers: $\text{wp}(t, \varphi)$ for sequential Rust, $\text{WP}((P, \sigma, i), \varphi)$ for async concurrency.